

Soumil Paranjpay

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Education

University of California, San Diego

Sep. 2024 – Dec. 2025

Master of Science in Electrical & Computer Engineering – CGPA: **3.78/4.00**

Vishwakarma Institute of Technology, Pune

Aug. 2020 – Jun. 2024

B. Tech in Electronics & Telecommunication Engineering – CPI: **8.89/10.00** (Class Rank: 5/307)

Work Experience

Apple – GPU Architecture Modelling Engineer

Cupertino, CA | Jan. 2026 – Present

- Led feature bring-up and debug for the GPU memory subsystem performance and functional models, with a focus on L2 cache and directory behavior.
- Collaborated closely with software, driver, and microarchitecture teams to diagnose issues and ensure correct system-level functionality.

JPMorganChase – Software Engineering Intern

Mumbai, India | Jan. 2024 – Jun. 2024

- Spearheaded the development of an **automation tool** for .NET Framework to Core migration, achieving a **95% reduction in migration times** through advanced code analysis.
- Collaborated with cross-functional teams to ensure seamless integration and adoption of the tool, significantly enhancing project efficiency.

JPMorganChase – Summer Intern

Bengaluru, India | Jun. 2023 – Jul. 2023

- Assisted in deployment, debugging, and writing Kubernetes manifests for the WMDM team in the document management space.

Projects

Out-of-Order RISC-V Processor (C++)

Jan. 2025 – Mar. 2025

- Architected and modelled a scalar Out-of-Order (OoO) processor incorporating **speculative execution** (GShare predictor), **dynamic instruction scheduling** via an instruction queue, and precise state management with **in-order retirement**.
- Developed a configurable C++ **performance simulator** to quantify the CPI impact of varying microarchitectural parameters (e.g., instruction queue depth, predictor table size) across different SPEC-CPU workloads.

Low-Power Dual Core Machine Learning Accelerator (Verilog)

Jan. 2025 – Mar. 2025

- Designed and optimized RTL for a 16x16 **systolic array** for attention calculation.
- Implemented multi-VT place-and-route, **clock gating**, **power gating** to reduce power and improve PPA metrics by **45%**.
- Optimized RTL for sparse vector multiplication and implemented dual-core communication using async **4-way handshake** protocol.

Skills

Languages, Tools, Platforms: C, C++, SystemC, Python, Metal Shading Language, Ruby, ARM ISA